

A.9 POST-HOC CALIBRATION STRATEGIES

Several post hoc strategies have been developed for calibrating the predictions of a model. These have the advantage of flexibility, as they operate only on the outputs of a model and do not require that any changes be made to the model itself. Some methods include:

- **Temperature scaling (TS)** (Guo et al., 2017) simply scales the logits by a temperature parameter $T > 1$ to smooth the predictions. The scaling parameter T can be tuned on a validation set.
- **Ensemble temperature scaling (ETS)** (Zhang et al., 2020) learns an ensemble of temperature-scaled predictions with uncalibrated predictions ($T = 1$) and uniform probabilistic outputs ($T = \infty$).
- **Vector scaling (VS)** (Guo et al., 2017) scales the entire output vector of class probabilities, rather than just the logits.
- **Multi-class isotonic regression (IRM)** (Zhang et al., 2020) is a multiclass generalization of the famous isotonic regression method (Zadrozny & Elkan, 2002): it ensembles predictions and labels, then learns a monotonically increasing function to map transformed predictions to labels.
- **Order-invariant calibration** (Rahimi et al., 2020) uses a neural network to learn an intra-order-preserving calibration function that can preserve a model’s top-k predictions.
- **Spline calibration** instead uses splines to fit the calibration function (Gupta et al., 2021).
- **Dirichlet calibration** (Kull et al., 2019) models the distribution of outputs using a Dirichlet distribution, using simple log-transformation of the uncalibrated probabilities which are then passed to a regularized fully connected neural network layer with softmax activation.

For node classification, some graph-specific post-hoc calibration methods have been proposed. **CaGCN** (Wang et al., 2021) uses the graph structure and an additional GCN to produce node-wise temperatures. **GATS** (Hsu et al., 2022) extends this idea by using graph attention to model the influence of neighbors’ temperatures when learning node-wise temperatures. We use the post hoc calibration baselines provided by Hsu et al. in our experiments.

All of the above methods, and others, may be applied to the output of any model including one using G- Δ UQ. As we have shown, applying such post hoc methods to the outputs of the calibrated models may improve uncertainty estimates even more. Notably, calibrated models are expected to produce confidence estimates that match the true probabilities of the classes being predicted (Naeini et al., 2015; Guo et al., 2017; Ovadia et al., 2019). While poorly calibrated CIs are over/under confident in their predictions, calibrated CIs are more trustworthy and can also improve performance on other safety-critical tasks which implicitly require reliable prediction probabilities (see Sec. 5). We report the top-1 label expected calibration error (ECE) (Kumar et al., 2019; Detlefsen et al., 2022). Formally, let p_i be the top-1 probability, c_i be the predicted confidence, b_i a uniformly sized bin in $[0, 1]$. Then,

$$ECE := \sum_i^N b_i \| (p_i - c_i) \|$$

A.10 DETAILS ON GENERALIZATION GAP PREDICTION

Accurate estimation of the expected generalization error on unlabeled datasets allows models with unacceptable performance to be pulled from production. To this end, generalization error predictors (GEPs) (Garg et al., 2022; Ng et al., 2022; Jiang et al., 2019; Trivedi et al., 2023a; Guillory et al., 2021) which assign sample-level scores, $S(x_i)$ which are then aggregated into dataset-level error estimates, have become popular. We use maximum softmax probability and a simple thresholding mechanism as the GEP (since we are interested in understanding the behavior of confidence indicators), and report the error between the predicted and true target dataset accuracy: $GEP_{Error} := ||\text{Acc}_{target} - \frac{1}{|\mathcal{X}|} \sum_i \mathbb{I}(S(\bar{x}_i; F) > \tau)||$ where τ is tuned by minimizing GEP error on the validation dataset. We use the confidences obtained by the different baselines as sample-level scores, $S(x_i)$ corresponding to the model’s expectation that a sample is correct. The MAE between the estimated error and true error is reported on both in- and out-of -distribution test splits provided by the GOOD benchmark.

A.11 RESULTS ON GENERALIZATION ERROR PREDICTION

GEP Experimental Setup. GEPs (Garg et al., 2022; Ng et al., 2022; Jiang et al., 2019; Trivedi et al., 2023a; Guillory et al., 2021) aggregate sample-level scores capturing a model’s uncertainty about the correctness of a prediction into dataset-level error estimates. Here, we use maximum softmax probability for scores and a thresholding mechanism as the GEP. (See Appendix A.10 for more details.) We consider READOUT anchoring with both pretrained and end-to-end training, and report the mean absolute error between the predicted and true target dataset accuracy on the OOD test split.

GEP Results. As shown in Table 16, both pretrained and end-to-end G- Δ UQ outperform the vanilla model on 7/8 datasets. Notably, we see that pretrained G- Δ UQ is particularly effective as it obtains the best performance across 6/8 datasets. This not only highlights its utility as a flexible, light-weight strategy for improving uncertainty estimates without sacrificing accuracy, but also emphasizes that importance of structure, in lieu of full stochasticity, when estimating GNN uncertainties.

Table 16: **GOOD-Datasets, Generalization Error Prediction Performance.** The MAE between the predicted and true test error on the OOD test split is reported. G- Δ UQ variants outperform vanilla models on 7/8 datasets (GOODMotif(Basis,Covariate) being the exception). Pretrained G- Δ UQ is particularly effective at this task as it achieves the best performance overall on 6/8 datasets. Promisingly, we see that regular G- Δ UQ improves performance over the vanilla model on 6/8 datasets (even if it is not the best overall). We further observe that performing generalization error prediction is more challenging under covariate shift than concept shift on the GOODCMNIST, GOODMotif(Basis) and GOODMotif(Size) datasets. On these datasets, the MAE is almost twice as large than their respective concept shift counterparts, across methods. GOODSST2 is the exception, where concept shift is in fact more challenging. To the best of our knowledge, we are the first to investigate generalization error prediction on GNN-based tasks under distribution shift. Understanding this behavior further is an interesting direction of future work.

Method	CMNIST (Color)		MotifLPE (Basis)		MotifLPE (Size)		SST2	
	Concept($\downarrow$)	Covariate ($\downarrow$)	Concept($\downarrow$)	Covariate($\downarrow$)	Concept($\downarrow$)	Covariate($\downarrow$)	Concept($\downarrow$)	Covariate($\downarrow$)
Vanilla	0.200 $\pm$ 0.009	0.510 $\pm$ 0.089	0.045 $\pm$ 0.003	0.570 $\pm$ 0.012	0.324 $\pm$ 0.018	0.537 $\pm$ 0.146	0.117 $\pm$ 0.006	0.056 $\pm$ 0.044
G- Δ UQ	0.190 $\pm$ 0.010	0.493 $\pm$ 0.072	0.023 $\pm$ 0.003	0.572 $\pm$ 0.019	0.317 $\pm$ 0.007	0.528 $\pm$ 0.189	0.124 $\pm$ 0.016	0.054 $\pm$ 0.043
Pretr. G- Δ UQ	0.192 $\pm$ 0.005	0.387 $\pm$ 0.048	0.018 $\pm$ 0.012	0.573 $\pm$ 0.004	0.307 $\pm$ 0.016	0.356 $\pm$ 0.143	0.114 $\pm$ 0.004	0.030 $\pm$ 0.026

A.12 ADDITIONAL STUDY ON PRETRAINED G- Δ UQ

For the datasets and data shifts on which we reported out-of-distribution calibration error of pretrained vs. in-training G- Δ UQ earlier in Fig. 4, we now report additional results for in-distribution and out-of-distribution accuracy as well as calibration error. We also include results for the additional GOODMotif-basis benchmark for completeness, noting that the methods provided by the original benchmark (Gui et al., 2022) generalized poorly to this split (which may be related to why G- Δ UQ methods offer little improvement over the vanilla model.) Fig. 8 shows these extended results. By these additional metrics, we again see the competitiveness of applying G- Δ UQ to a pretrained model versus using it in end-to-end training.

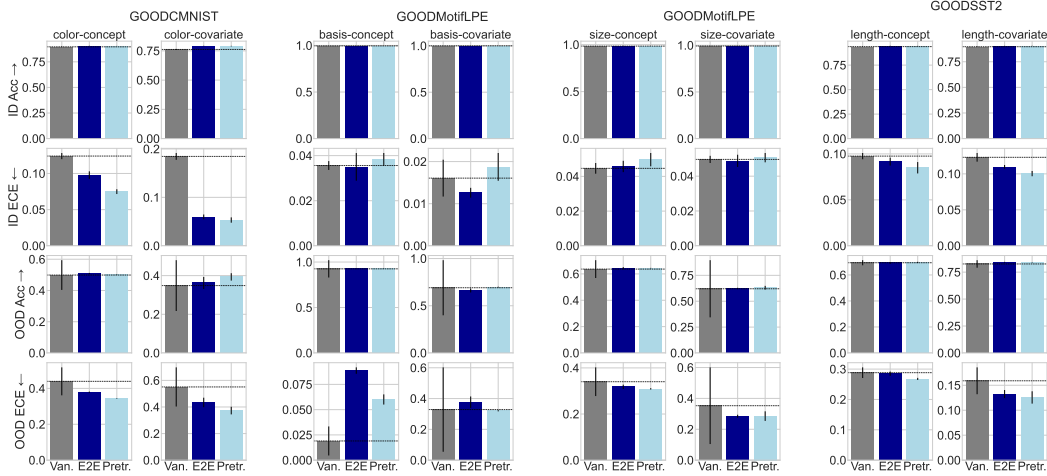


Figure 8: Evaluating Pretrained G- Δ UQ. Here, we report the performance of pretrained G- Δ UQ models vs. end-to-end and vanilla models with respect to in-distribution and out-of-distribution accuracy as well as expected calibration error. With the exception of the GOODMotif (basis) dataset, pretrained G- Δ UQ improves the OOD ECE over both the vanilla model and end-to-end G- Δ UQ at comparable or improved OOD accuracy on 7/8 datasets. Furthermore, pretrained G- Δ UQ also improves the ID ECE on all but the GOODMotif (size) datasets (6/8), where it performs comparably to the vanilla model, and maintains the ID accuracy. (We note that all methods are comparably better calibrated on the GOODMotif ID data than GOODCMNIST/GOODSST2 ID data; we suspect this is because there may exist simple shortcuts available in the GOODMotif dataset that can be used on the ID test set effectively.) Overall, these results clearly demonstrate that pretrained G- Δ UQ does offer some performance advantages over end-to-end G- Δ UQ and does so at reduced training times (see Table. A.13). For example, on GOODCMNIST (covariate shift), pretrained G- Δ UQ is not only 50% faster than end-to-end G- Δ UQ, it also improves OOD accuracy and OOD ECE over both the vanilla and end-to-end G- Δ UQ models.